

5.5 Decomposing the Decline in Junior Employment into Flows

The decline in junior employment in adopting firms can arise through three channels: (i) reduced hiring, (ii) increased separations, or (iii) increased promotions to senior positions. Our LinkedIn-based résumé data can be viewed as a detailed matched employer-employee dataset, allowing us to track these worker flows over time. Specifically, we estimate separate DiD regressions of the following form:

$$y_{it} = \alpha + \beta (\text{Adopt}_i \times \text{Post}_t) + \delta_t + \gamma_i + \xi_{pt} + \varepsilon_{it}, \quad (15)$$

where y_{it} denotes the number of junior hires, number of junior separations, or the junior promotion rate³¹ in firm i at quarter t . γ_i denotes firm fixed effects, δ_t are time fixed effects and ξ_{pt} are industry-by-time fixed effects.

Table 4 reports the estimated β from Equation 15. The results indicate that the sharp contraction in junior employment among adopters is driven primarily by a slowdown in hiring. Specifically, the coefficient on *Hiring* implies that, relative to non-adopters, GenAI-adopting firms hired on average 4.0 fewer junior workers per quarter after 2023Q1—about an 80 percent reduction relative to the pre-period mean of 5.06 hires per firm-quarter. Junior separations also declined among adopters, by roughly 1.1 workers per quarter, partially offsetting the hiring contraction; but this reduction is about one-fourth the magnitude of the hiring decline, so net junior employment still contracts sharply.³² Junior promotion rates rose by a small but statistically significant 0.033 percentage points (relative to a pre-period mean of 0.82 percent). Because promotions move workers out of the junior pool, this rise further reduces the junior stock, though its magnitude is very small relative to the hiring channel. Appendix A.28 explores sectoral heterogeneity in the decline in junior hiring, showing that the contraction is broad-based across industries and not driven disproportionately by any single sector.

³¹The junior promotion rate is defined as $\frac{\text{Promotions}_t}{\text{Junior Employment}_{t-1}}$, expressed in percentage points.

³²See Supplemental Appendix A.26 for the time-series DiD results for junior hires and exits.